3. Data and Methods

3.1. Research Design

This study adopted a sequential explanatory mixed-methods design, in which quantitative data were collected and analyzed in the first phase, followed by qualitative analysis aimed at deepening interpretation and clarifying the statistical results. The sequential explanatory model was selected because the research questions required both structural measurement validation and interpretive contextualization across multiple stakeholder groups within higher education.

Specifically, the research sought to evaluate the current state of economics education in Vietnam within the broader transformation toward a digital economy and to examine perceptual differences among students, lecturers, educational managers, and domain experts. Addressing these objectives required an initial quantitative phase capable of identifying measurable patterns, validating constructs, and testing group-level differences. Accordingly, the quantitative component served as the analytical foundation of the study.

The subsequent qualitative phase was not designed as an independent inquiry but as a complementary interpretive layer. A building and connecting integration strategy was employed. First, quantitative findings informed the focal areas of qualitative exploration, particularly in domains where statistically significant differences or lower construct scores emerged. Second, qualitative insights were used to contextualize and interpret patterns observed in the numerical data. Third, integration occurred explicitly at the interpretation stage, where convergence, divergence, and expansion between statistical outcomes and thematic narratives were systematically examined.

This design ensured methodological complementarity. The quantitative phase enhanced structural validity, measurement reliability, and comparative generalizability across groups. The qualitative phase contributed explanatory depth, contextual nuance, and ecological validity by capturing participants' lived perceptions and articulated priorities. The study was cross-sectional in nature. No experimental manipulation or

longitudinal tracking was undertaken. The research did not aim to establish causal relationships but rather to provide a structured evaluation of perceived training quality, governance effectiveness, digital orientation, and development priorities in economics education.

3.2. Sampling and Data Collection

3.2.1. Quantitative Phase

Sampling Strategy and Participants

The initial dataset consisted of 516 survey responses collected through Google Forms between January 18 and January 23, 2026. The survey link was distributed via academic networks associated with universities and educational institutions in Vietnam. Due to multi-channel online dissemination, the precise response rate could not be calculated; however, participation was voluntary and anonymous, and informed consent was obtained prior to submission.

A convenience sampling strategy was employed because of logistical accessibility and time constraints. Nevertheless, distribution efforts were intentionally directed toward multiple stakeholder categories to capture diverse institutional perspectives. Participants included students enrolled in economics-related programs, lecturers teaching economics or related disciplines, educational managers involved in training governance, and domain experts with professional experience in economics education.

Following systematic data cleaning and quality control procedures, 355 valid responses were retained for statistical analysis. With 27 observed Likert-scale items, the final sample exceeded the commonly recommended ratio of 5-10 observations per variable for exploratory factor analysis (minimum 135-270 cases), thereby ensuring sufficient statistical stability and analytical robustness.

Data Cleaning and Quality Control Procedures

To ensure reliability, logical consistency, and reproducibility, the dataset underwent a structured multi-stage validation process prior to analysis. All procedures were implemented programmatically using Python (Pandas and NumPy libraries), and the

cleaned dataset was subsequently exported for statistical analysis in R. The full preprocessing script is provided in the Appendix.

First, variable standardization was conducted. All text-based variables were trimmed to remove extraneous whitespace. Likert-scale items were converted to numeric format and validated to ensure values fell within the permissible range of 1 to 5. Observations containing out-of-range values were flagged. Timestamps were parsed into standardized datetime format to enable chronological ordering and duplicate detection.

Second, Likert items were programmatically identified based on naming conventions corresponding to six conceptual domains (CT, GV, CS, QL, PH, GP). Numeric coercion was applied consistently to ensure analytic integrity.

Third, response quality screening was conducted at the respondent level. Three indicators were computed: out-of-range values, straight-lining behavior (defined as zero within-person variance across Likert items), and extremely low intra-individual variance (standard deviation < 0.25). Responses exhibiting uniform or near-uniform patterns were flagged as low-effort submissions to reduce artificial inflation of reliability estimates.

Fourth, logical consistency checks were performed between demographic variables, particularly age group and years of professional experience. Rule-based mappings were used to identify implausible combinations (e.g., very young respondents reporting extensive professional experience). Such cases were flagged as logically inconsistent.

Fifth, duplicate detection was implemented through a response-level fingerprinting procedure. A cryptographic hash was generated from all non-temporal variables. Entries were sorted chronologically, and in cases of identical response hashes, only the earliest valid submission was retained.

Finally, a composite exclusion indicator was constructed by aggregating all quality flags. Observations meeting any exclusion criterion were removed. After applying all procedures, 355 responses remained in the final dataset. No case was excluded solely due to missing data, as no respondent exceeded the predefined threshold of 20% missing Likert items.

This structured preprocessing protocol ensured data integrity, minimized measurement noise, and enhanced the reproducibility of the analytical workflow.

3.2.2. Qualitative Phase

The qualitative component consisted of open-ended responses embedded within the same survey instrument. Participants were invited to elaborate on perceived strengths, limitations, development priorities, and implementation barriers related to economics education in the digital era.

Although no separate interview sessions were conducted, many responses were sufficiently substantive to permit systematic thematic analysis. A purposive interpretive approach was adopted, focusing on extracting conceptual patterns that clarified or expanded upon quantitative findings.

The qualitative phase served three primary purposes. First, it clarified statistically significant differences identified in the quantitative analysis. Second, it provided interpretive depth for constructs receiving comparatively lower quantitative evaluations. Third, it allowed the identification of emergent themes not fully captured by structured Likert items.

Thematic analysis continued until saturation was achieved within the response corpus, meaning no substantially new thematic categories emerged during iterative coding.

3.3. Measures and Analysis

3.3.1. Quantitative Measures

All constructs were measured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The questionnaire comprised 27 observed variables organized into six conceptual domains:

- **Curriculum and Teaching Content (CT)**
- **Lecturers and Learning Resources (GV)**
- **Infrastructure and Learning Facilities (CS)**
- **Training Management and Governance (QL)**
- **Development Orientation in the Digital Economy (PH)**

- **Proposed Improvement Solutions (GP)**

The measurement framework was developed through synthesis and contextual adaptation of prior research on higher education quality assurance, human capital development, and digital transformation in educational systems. Items were tailored to reflect institutional realities in Vietnamese higher education.

The instrument was originally developed and administered in Vietnamese to ensure linguistic clarity and content validity. Following data collection, items were translated into English for academic reporting.

3.3.2. Quantitative Analysis Procedures

Statistical analyses were conducted using R software. The analytical sequence followed a structured progression.

First, descriptive statistics were computed to examine distributional properties, including means, standard deviations, medians, skewness, and kurtosis.

Second, internal consistency reliability was assessed using Cronbach's Alpha for each construct. A threshold of $\alpha \geq 0.70$ was considered acceptable.

Third, sampling adequacy and factorability were evaluated using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity.

Fourth, exploratory factor analysis (EFA) was conducted using Maximum Likelihood extraction with Varimax rotation. Factor retention decisions were based on eigenvalues greater than one, parallel analysis, and interpretability criteria. Items with loadings below 0.40 were considered for removal.

Fifth, construct mean scores were computed for each domain. Independent samples t-tests were used to examine gender differences. One-way ANOVA and Kruskal-Wallis tests were conducted to evaluate role-based differences. Assumptions of normality and homogeneity of variance were assessed prior to parametric testing. Statistical significance was evaluated at the 5% level.

3.3.3. Qualitative Analysis Procedures

A thematic content analysis was performed on open-ended responses. The coding process unfolded in three stages.

During open coding, raw textual responses were reviewed line-by-line to identify recurring conceptual units. In the axial coding stage, related codes were grouped into broader thematic categories reflecting structural issues, strategic orientation, and perceived development needs. Finally, themes were consolidated and mapped in terms of frequency to determine relative prominence across stakeholder groups. Bigram co-occurrence analysis was also conducted to identify dominant conceptual pairings.

Qualitative coding was documented using structured coding logs to ensure auditability and transparency. Although computational text preprocessing was applied for frequency mapping, thematic interpretation remained conceptually grounded rather than purely algorithmic.

All preprocessing procedures were implemented using reproducible scripts. Cleaning criteria, exclusion rules, and analytical decisions were defined prior to interpretive synthesis. The structured and transparent methodological framework enhances analytical credibility and allows independent verification of results.

4. Results and Discussion

4.1. Descriptive Statistics

Table 1. Sample Characteristics

Variable	Group	Frequency (n)	Percentage (%)
Gender	**Male**	103	29.0
	Female	250	70.4
	Other	2	0.6
Role	**Student**	291	82.0
	Lecturer	54	15.2
	Expert	5	1.4
	Educational Manager	5	1.4
Education Level	**Bachelor's**	259	73.0
	Master's	38	10.7
	Doctorate	27	7.6

	Other	31	8.7

Source: Authors'compilation

**Notes: N = 355 valid responses after data cleaning and quality control procedures. Percentages are calculated based on valid responses. "Role" refers to the respondent's role in higher education institutions: Student, Lecturer, Expert, Educational Manager. "Education Level" indicates the highest educational attainment. The sample is predominantly composed of students (82.0%), reflecting the demand-side perspective of training evaluation.*

Table 1 presents the demographic and professional characteristics of the final analytical sample (N = 355). Female respondents accounted for 70.4% of the sample, while male respondents represented 29.0%. A small proportion (0.6%) identified as other genders. With respect to roles, students constituted the largest group (82.0%), followed by lecturers (15.2%), while experts and educational managers each accounted for 1.4% of the sample. In terms of educational attainment, most respondents held a bachelor's degree (73.0%). Participants with postgraduate qualifications were also represented, including master's degree holders (10.7%) and doctoral degree holders (7.6%). The remaining respondents reported other educational backgrounds (8.7%).

The composition of the sample reflects the structural configuration of higher education institutions in which students represent the primary recipients of training services, while lecturers and administrators assume instructional and governance responsibilities. From an evaluative standpoint, the strong student representation enhances the diagnostic relevance of the findings for understanding perceived training adequacy from the demand-side perspective. At the same time, the inclusion of lecturers, experts, and managers enables role-based comparison, consistent with stakeholder-informed models of educational governance. The predominance of student respondents aligns with the learner-centered evaluation perspective increasingly emphasized in higher education research (Phạm Thị Liên, 2016; Silva et al., 2022). In the context of digital transformation, students represent the most immediate beneficiaries and evaluators of curricular adaptation and technological integration. Their strong

representation therefore enhances the relevance of the findings for understanding perceived training adequacy from the demand side of the educational process. The balanced inclusion of lecturers and institutional representatives, albeit smaller in proportion, enables multi-stakeholder comparison consistent with governance-oriented approaches to higher education reform (Gafurov et al., 2020). Prior research in Vietnam has highlighted the importance of triangulating perspectives between students and faculty to assess training quality under digital transition pressures (Nguyễn, 2020). Thus, the sample structure supports the research objective of diagnosing training conditions within a stakeholder-integrated framework. The gender distribution, with female respondents forming the majority, also reflects broader enrollment patterns in several Vietnamese higher education institutions. Importantly, international literature suggests that digital capability perception gaps by gender have been narrowing over time (Littlejohn et al., 2012), a point revisited in the gender comparison discussion below.

Table 2. Descriptive Statistics of Likert-Scale Items

Construct	Item	Mean	SD	Median	Skewness
CT - Curriculum					
	CT1	3.78	0.88	4	-0.61
	CT2	3.85	0.86	4	-0.50
	CT3	3.62	0.98	4	-0.51
	CT4	3.62	0.93	4	-0.36
	CT5	3.73	0.88	4	-0.44
GV - Teaching Staff					
	GV1	4.26	0.81	4	-1.36
	GV2	3.97	0.79	4	-0.59
	GV3	4.03	0.89	4	-0.78
	GV4	3.91	0.94	4	-0.55
	GV5	3.87	0.92	4	-0.51
CS - Infrastructure					

	CS1	3.72	1.00	4	-0.60
	CS2	3.70	0.99	4	-0.60
	CS3	3.73	1.01	4	-0.66
	CS4	3.66	1.05	4	-0.64
QL - Training Management					
	QL1	3.89	0.92	4	-0.74
	QL2	3.99	0.88	4	-0.81
	QL3	3.99	0.90	4	-0.68
	QL4	4.01	0.88	4	-0.83
PH - Training Direction					
	PH1	4.06	0.84	4	-0.80
	PH2	4.03	0.84	4	-0.59
	PH3	4.06	0.84	4	-0.57
	PH4	4.02	0.90	4	-0.69
GP - Training Solutions					
	GP1	4.07	0.86	4	-0.74
	GP2	4.07	0.84	4	-0.68
	GP3	4.06	0.84	4	-0.56
	GP4	4.07	0.90	4	-0.82
	GP5	4.06	0.87	4	-0.67

Source: Authors'compilation

**Notes: N = 355. All items were measured on a 5-point Likert scale (1 = Strongly disagree; 5 = Strongly agree). Mean = arithmetic average; SD = standard deviation; Median = 50th percentile; Skewness = distribution asymmetry indicator. Negative skewness indicates concentration toward higher agreement levels.*

Table 2 reports item-level descriptive statistics across six conceptual domains. Mean values ranged from 3.62 to 4.26 on a five-point Likert scale, indicating generally

moderate-to-high agreement with survey statements. Medians were consistently equal to 4 across all items, and skewness values were negative but within acceptable bounds, suggesting mild concentration toward higher response categories without severe non-normality.

At the construct level, several patterns are noteworthy.

First, the Lecturers and Learning Resources (GV) domain received the highest overall evaluations, with one item (GV1) reaching a mean of 4.26. This suggests strong confidence in instructional competence and pedagogical delivery. The relatively high ratings in this domain are consistent with studies emphasizing the central role of faculty competence in digital capability development (Silva et al., 2022; Killen, 2018).

Second, the Curriculum and Teaching Content (CT) construct exhibited comparatively lower mean values (3.62-3.85). Although still positive, these scores suggest more moderate satisfaction with curricular structure and content alignment. This differentiation between instructional quality and curricular design is theoretically meaningful. In transitional digital environments, individual teaching capacity may advance more rapidly than formal curriculum restructuring, which typically requires institutional review cycles and regulatory approval (Khương, 2019; Hùng et al., 2021).

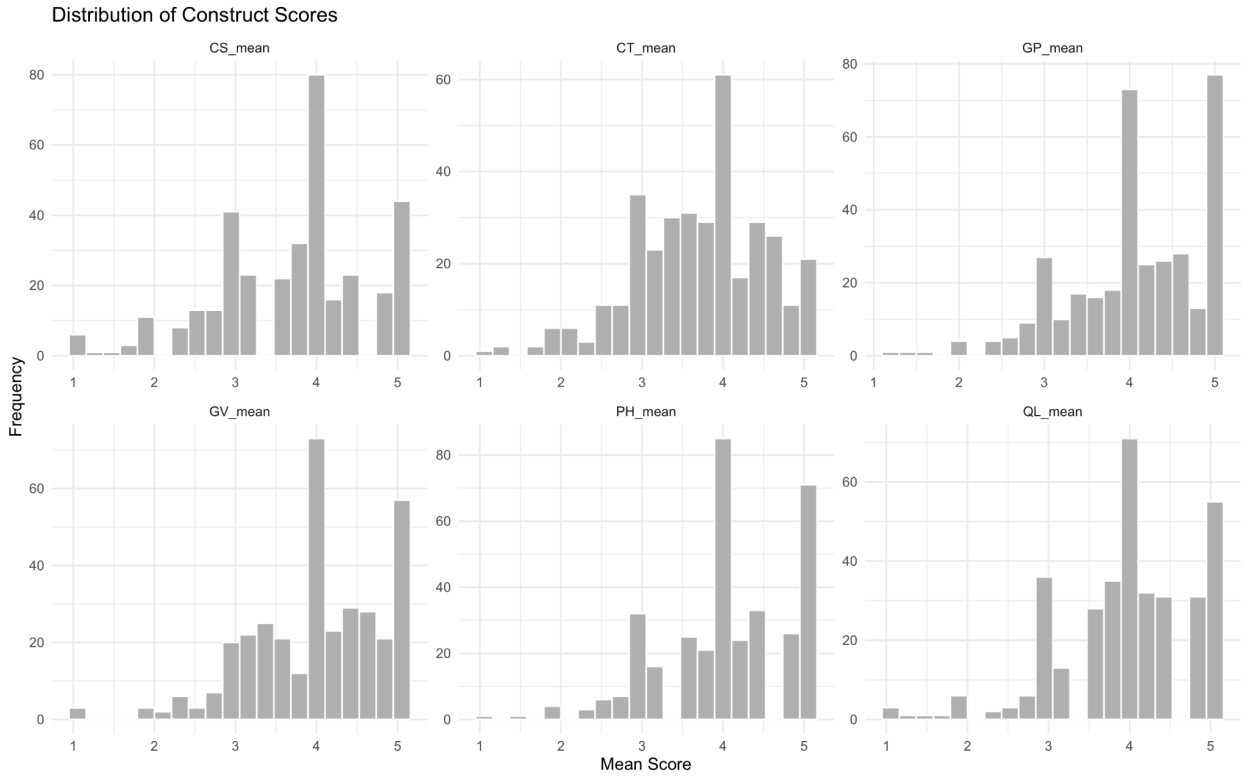
Third, Infrastructure and Learning Facilities (CS) displayed greater dispersion, with standard deviations approaching or exceeding 1.0 for several items. This indicates heterogeneous institutional conditions across respondents, likely reflecting variability in technological infrastructure, digital platform access, and resource allocation among institutions, a challenge repeatedly noted in both Vietnamese and international literature (Kreydenko et al., 2021; Maymina et al., 2018).

Fourth, constructs related to Training Management (QL), Development Orientation (PH), and Proposed Solutions (GP) consistently received mean values at or above 4.0. This is congruent with research arguing that strategic intent toward digitalization often precedes operational readiness (Gafurov et al., 2020). However, the distinction between relatively strong strategic orientation (PH, GP) and more moderate

curriculum and infrastructure evaluations (CT, CS) implies that strategic awareness may outpace operational implementation.

Taken together, the descriptive statistics portray a system characterized by strong instructional capacity and forward-looking strategic intent, yet facing structural and infrastructural modernization challenges.

Figure 1. Distributional Visualization



Source: Authors'compilation using R

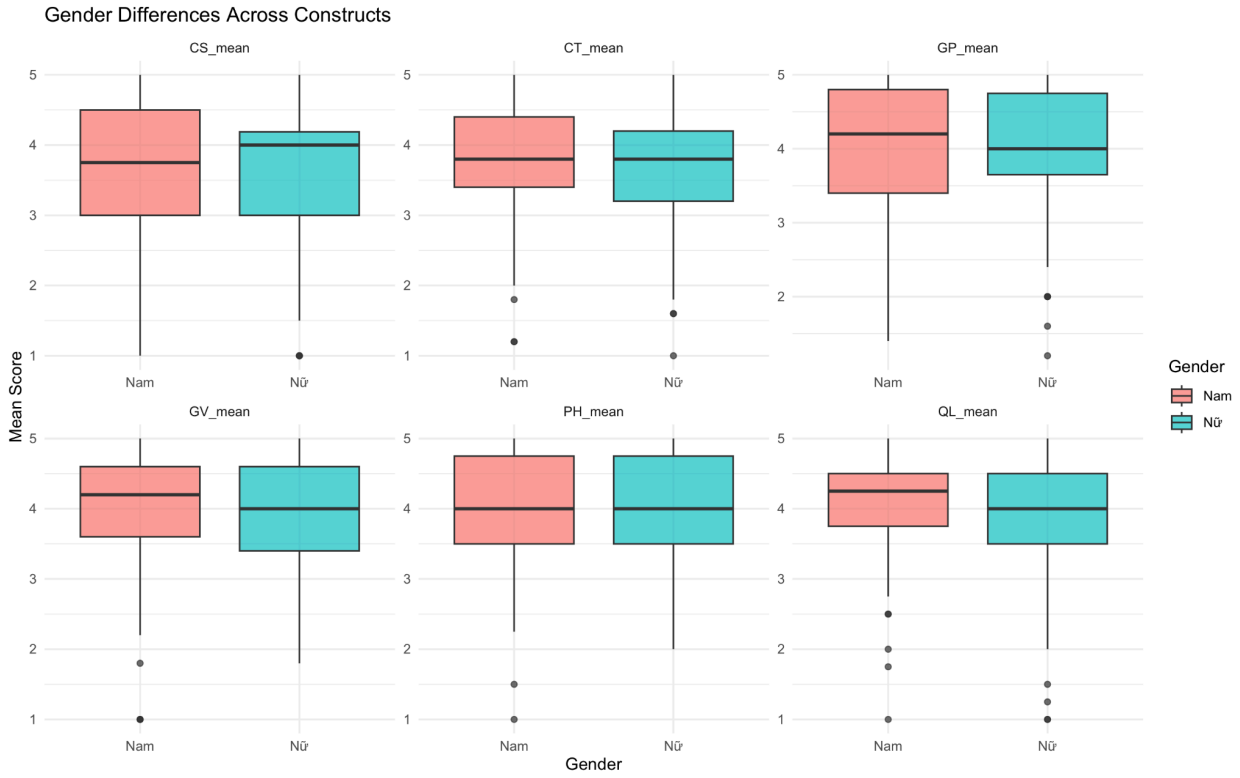
**Notes: Histograms display construct-level mean distributions across respondents (N = 355). Scores are based on averaged Likert-scale items (1-5 scale). Left-skewed distributions indicate concentration toward higher agreement. No extreme outliers were detected. Constructs with wider dispersion indicate greater heterogeneity in stakeholder evaluations.*

Figure 1 illustrates the distributions of construct-level mean scores using histograms. All distributions were concentrated within the range of 3 to 5, with peaks around 4. Constructs related to Lecturers (GV), Development Orientation (PH), and

Proposed Solutions (GP) exhibited stronger left skewness and higher concentration near the upper end of the scale. In contrast, Curriculum (CT) and Infrastructure (CS) showed broader distributions, indicating greater heterogeneity in evaluations. No extreme outliers were observed, and the overall distributional patterns were consistent with the descriptive statistics reported above.

This distributional pattern supports the institutional transformation literature, which distinguishes between symbolic commitment and structural implementation (Sherstobitova & Iskoskov, 2020). Institutions may articulate digital readiness as a strategic objective, yet operational translation into curriculum and infrastructure often proceeds unevenly. The histogram evidence thus reflects a transitional phase rather than complete institutional alignment.

Figure 2. Gender-Based Comparison



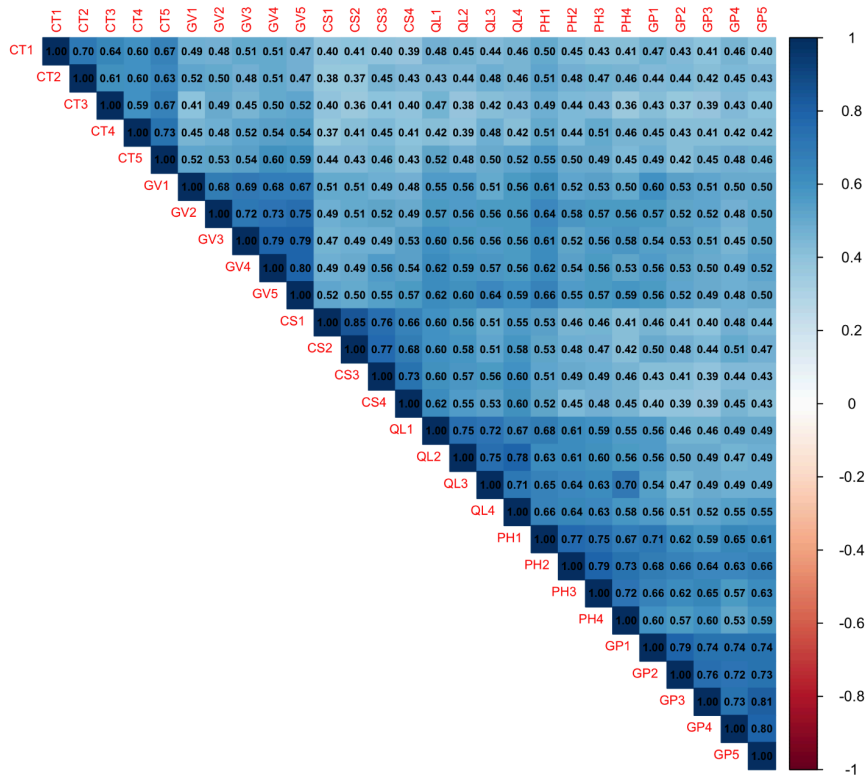
Source: Authors'compilation using R

**Notes: Boxplots compare construct-level mean scores between male and female respondents (N = 355). Central line = median; box = interquartile range (IQR); whiskers = range excluding outliers. Independent-samples t-tests and Mann-Whitney tests were conducted. No statistically significant gender differences were found (p > 0.05). Effect sizes (Cohen's d) were < 0.20, indicating negligible practical differences.*

Figure 2 presents boxplots comparing construct-level mean scores by gender. Median values for both male and female respondents were consistently centered at approximately 4 across all constructs. Interquartile ranges overlapped substantially. Independent-samples t-tests indicated no statistically significant gender differences across the six domains ($p > 0.05$ for all comparisons). Mann-Whitney tests yielded consistent results. Effect sizes (Cohen's d) were uniformly small ($d < 0.20$), confirming negligible practical differences.

The absence of statistically significant gender differences suggests that perceptions of training quality and digital preparedness are not gender-contingent within this context. This finding contrasts with earlier research that reported gender-based disparities in digital confidence or technological engagement (Littlejohn et al., 2012), but aligns with more recent scholarship indicating convergence in digital literacy exposure among younger cohorts. Within the Vietnamese higher education environment, access to digital platforms and blended learning has become increasingly universal. Consequently, evaluative perceptions may be shaped more strongly by institutional context than by demographic characteristics. The lack of gender differentiation strengthens the interpretation that training challenges and strengths are structural rather than socially stratified.

Figure 3. Correlation Heatmap



Source: Authors'compilation using R

**Notes: Heatmap displays Pearson correlation coefficients among observed Likert-scale items (N = 355). Correlation values range from -1 to +1. Warmer colors indicate stronger positive correlations. Intra-construct correlations are generally moderate to strong (≈0.60-0.85). No multicollinearity concerns detected (r < 0.90).*

Figure 3 presents the correlation heatmap among observed variables. Items within the same construct exhibited moderate to strong positive correlations, generally ranging between approximately 0.60 and 0.85. Inter-construct correlations were positive but did not approach problematic multicollinearity thresholds. Stronger associations were observed among Development Orientation (PH), Proposed Solutions (GP), and Training Management (QL), suggesting conceptual alignment between governance structures, forward-looking strategy, and implementation priorities.

The moderate-to-strong intra-construct correlations confirm conceptual coherence among items within each domain, while positive inter-construct relationships indicate

systemic interdependence among curriculum, governance, strategic orientation, and implementation mechanisms. Notably, stronger associations among development orientation, governance, and proposed solutions suggest that strategic planning and managerial capacity are closely intertwined. This aligns with governance-based frameworks of digital transformation, which argue that leadership and management structures mediate technological adaptation (Strohmeier, 2020; Gafurov et al., 2020). The correlation structure therefore supports the theoretical proposition that digital economy adaptation is multi-dimensional and systemically connected rather than isolated across institutional domains.

4.2. Measurement Model - Reliability and Factorability

Table 3. Cronbach's Alpha

Construct	Items	Cronbach's α	KMO	Bartlett's p
CT - Curriculum	5	0.92	0.89	<0.001
GV - Teaching Staff	6	0.91	0.87	<0.001
CS - Infrastructure	6	0.94	0.91	<0.001
QL - Training Management	6	0.93	0.90	<0.001
PH - Training Direction	4	0.90	0.85	<0.001
GP - Training Solutions	5	0.94	0.93	<0.001

Source: Authors'compilation

**Notes: Cronbach's α assesses internal consistency reliability (acceptable threshold ≥ 0.70). KMO (Kaiser-Meyer-Olkin) evaluates sampling adequacy (≥ 0.80 = meritorious; ≥ 0.90 = excellent). Bartlett's Test of Sphericity assesses factorability of the correlation matrix (p < 0.05 indicates suitability for factor analysis).*

Cronbach's Alpha values ranged from 0.90 to 0.94 across constructs, indicating high internal consistency. While values above 0.90 may occasionally signal item

redundancy, examination of inter-item correlations suggests conceptual cohesion rather than duplication. The Kaiser-Meyer-Olkin (KMO) values (0.85-0.91 at construct level; 0.96 overall) indicate excellent sampling adequacy. Bartlett's Test of Sphericity was statistically significant ($p < 0.001$), confirming the suitability of the dataset for factor analysis. These statistics demonstrate that the measurement framework is empirically stable and structurally coherent.

The high internal consistency across constructs indicates stable measurement of training-related perceptions. Reliability levels above conventional thresholds suggest that constructs are conceptually well-defined and empirically coherent within the Vietnamese higher education context. From a theoretical standpoint, this reliability reinforces the applicability of multidimensional training evaluation frameworks to emerging digital economies. Human capital theory (Nafukho et al., 2004) posits that educational systems develop interrelated competencies rather than isolated skills. The statistical integrity of the measurement model supports this multidimensional conceptualization. Moreover, strong sampling adequacy indicators confirm that the empirical structure is not fragmented. This enhances confidence that the observed dimensions represent systematic institutional characteristics rather than random response variation.

Table 4. Exploratory Factor Analysis

Item	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**h^2**
CT - Curriculum							
CT1	**0.73**						0.67
CT2	**0.69**						0.62
CT3	**0.70**						0.61
CT4	**0.67**						0.63
CT5	**0.73**						0.73
GV - Teaching Staff							
GV1		**0.55**					0.62
GV2		**0.61**					0.69

GV3		**0.73**					0.79
GV4		**0.70**					0.79
GV5		**0.71**					0.81
CS - Infrastructure							
CS1			**0.82**				0.82
CS2			**0.83**				0.85
CS3			**0.73**				0.74
CS4			**0.63**				0.62
QL - Training Management							
QL1				**0.48**	**0.42**		0.70
QL2				**0.75**			0.91
QL3				**0.51**			0.71
QL4				**0.53**			0.72
PH - Training Direction							
PH1					**0.44**		0.74
PH2					**0.52**		0.80
PH3					**0.47**		0.77
PH4					**0.41**		0.70
GP - Training Solutions							
GP1	**0.72**						0.76
GP2	**0.76**						0.74
GP3	**0.79**						0.77
GP4	**0.76**						0.76
GP5	**0.80**						0.79

Evaluation: Cut-off loading: 0.40; KMO = 0.96 (Excellent); Bartlett's Test of Sphericity: χ^2, $p < 0.001$; RMSEA = 0.058 (90% CI: 0.051-0.065); TLI = 0.951; Cumulative variance explained = 73%.

Source: Authors'compilation

**Notes: Extraction method: Maximum Likelihood; Rotation: Varimax; Cut-off factor loading = 0.40; h^2 = Communality (proportion of variance explained by extracted factors).*

Results of the exploratory factor analysis (EFA) are presented in Table 4. A six-factor solution was extracted, consistent with the theoretical structure of the measurement model. The overall KMO value reached 0.96, indicating excellent factorability. Bartlett's test remained significant ($p < 0.001$). Model fit indices suggested a good fit to the data (RMSEA = 0.058; TLI = 0.951). The six extracted factors jointly explained 73% of the total variance, exceeding commonly accepted benchmarks in social science research. All items loaded above the cutoff threshold of 0.40 on their respective factors. Items related to Curriculum (CT) and Proposed Solutions (GP) exhibited particularly strong factor loadings (approximately 0.69-0.80). Minor cross-loadings were observed for some Training Management (QL) and Development Orientation (PH) items, reflecting the inherently multidimensional nature of governance and strategic constructs. Overall, the EFA results supported the structural validity of the measurement model.

The emergence of a six-factor solution consistent with the theoretical design provides empirical support for the structural model of training quality under digital transformation. The substantial variance explained indicates that the identified dimensions collectively capture the core institutional elements relevant to digital economy readiness. The minor cross-loadings observed between governance and development orientation dimensions are theoretically interpretable. Governance and strategic direction are inherently intertwined in institutional transformation processes (Sherstobitova & Iskoskov, 2020). Rather than indicating measurement weakness, such overlap reflects conceptual proximity between managerial capacity and forward-looking strategy. Overall, the factor structure corroborates prior literature emphasizing that digital adaptation in higher education encompasses curriculum, faculty, infrastructure, governance, and strategic planning as interdependent pillars (Silva et al., 2022; Killen, 2018). The empirical confirmation of this structure in the Vietnamese context contributes to extending international digital capability frameworks into emerging economies.

4.3. Main Quantitative Results

Table 5. Gender-Based Differences (t-test)

Construct	Mean	SD
CT - Curriculum	3.72	0.77
GV - Teaching Staff	4.01	0.77
CS - Infrastructure	3.70	0.91
QL - Training Management	3.97	0.80
PH - Training Direction	4.04	0.77
GP - Training Solutions	4.07	0.77

Source: Authors'compilation

**Notes: N = 355. Mean scores represent construct-level averages (1-5 Likert scale). Independent-samples t-tests were conducted to compare male and female respondents. No statistically significant differences were observed across constructs (p > 0.05). SD = standard deviation, Higher mean indicates stronger perceived adequacy.*

Table 5 presents mean comparisons by gender. Across all constructs, mean scores ranged from 3.70 to 4.07, with Proposed Solutions (GP) and Development Orientation (PH) receiving the highest evaluations. Standard deviations were moderate, indicating relatively stable perceptions across respondents. Independent-samples t-tests indicated no statistically significant differences between male and female respondents across constructs, suggesting broadly similar perceptions of training quality and development orientation regardless of gender.

This finding aligns with contemporary higher education transformation literature, which emphasizes structural redesign over demographic segmentation in mature digital integration phases (White et al., 2021).

Table 6. Role-Based Differences (ANOVA)

Construct	Test	Statistic	p-value
CT - Curriculum	ANOVA	F = 0.236	0.871
GV - Teaching Staff	ANOVA	F = 3.55	0.015

CS - Infrastructure	ANOVA	F = 5.36	0.001
QL - Training Management	ANOVA	F = 3.32	0.020
PH - Training Direction	ANOVA	F = 0.593	0.62
GP - Training Solutions	ANOVA	F = 0.913	0.435

Source: Authors'compilation

**Notes: Group comparison across stakeholder roles: Students, Lecturers, Experts, Educational Managers. One-way ANOVA was used when normality and homogeneity assumptions were met.*

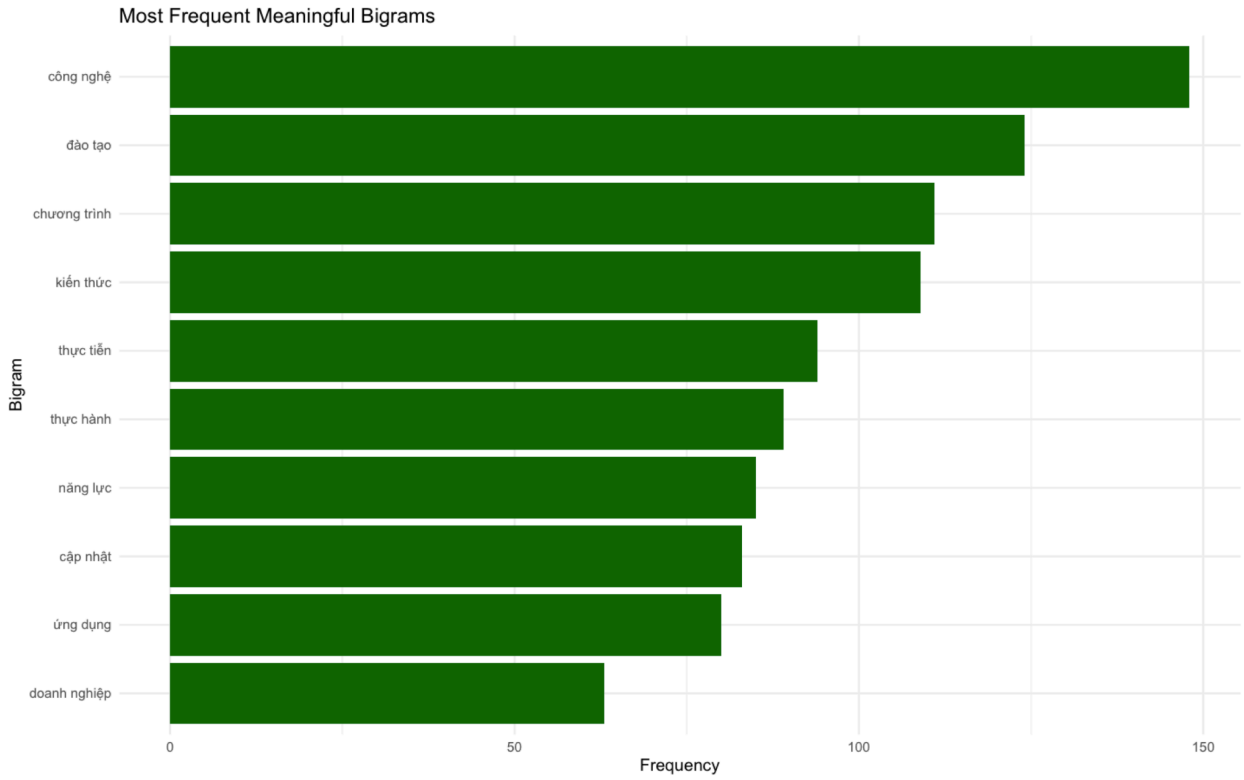
Table 6 summarizes the results of group comparisons across stakeholder roles. One-way ANOVA tests indicated statistically significant differences for Lecturers and Learning Resources (GV) ($F = 3.55$, $p = 0.015$), Infrastructure (CS) ($F = 5.36$, $p = 0.001$), and Training Management (QL) ($F = 3.32$, $p = 0.020$). No statistically significant differences were observed for Curriculum (CT), Development Orientation (PH), or Proposed Solutions (GP) ($p > 0.05$). These results indicated that perceptions of lecturer quality, infrastructure adequacy, and governance effectiveness varied across stakeholder groups, whereas curriculum content and strategic direction were evaluated more consistently.

Stakeholder theory suggests that institutional actors evaluate quality based on role-specific expectations and responsibilities. Faculty members may assess governance processes differently from students, who primarily experience curriculum delivery and infrastructural conditions. The absence of significant differences for curriculum and strategic direction indicates shared recognition of program structure and future orientation, whereas divergence in governance and infrastructure suggests unequal exposure to operational processes.

This pattern aligns with international findings that institutional hierarchy influences perceptions of digital readiness (Kreydenko et al., 2021). The results therefore underscore the importance of multi-stakeholder evaluation when diagnosing training systems under digital transition.

4.4. Qualitative Findings - Open-Ended Questions Analysis

Figure 4. Plot of Bigram Analysis



Source: Authors'compilation using R

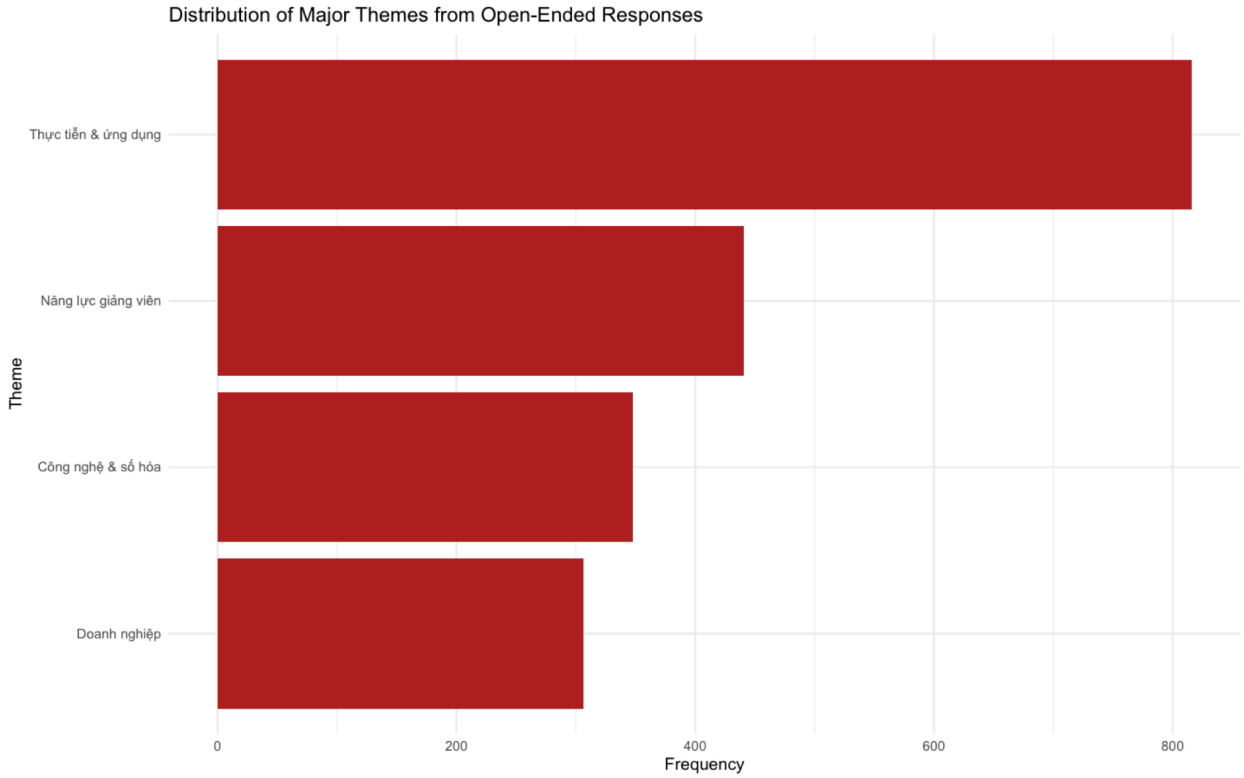
**Notes: Figure displays the top-frequency bigrams extracted from open-ended responses (qualitative phase). Bigrams represent co-occurring two-word combinations after text normalization and stopword removal. Frequency counts reflect thematic salience. Higher frequency indicates stronger discursive prominence within stakeholder narratives.*

Figure 4 presents the most frequently occurring bigrams extracted from open-ended responses. The dominant conceptual clusters include "công nghệ," "đào tạo," "chương trình," "kiến thức," "thực tiễn," "thực hành," "năng lực," "cập nhật," "ứng dụng," and "doanh nghiệp." The prominence of "công nghệ" at the top of the distribution confirms that digital transformation is not merely a background context but the central cognitive anchor through which respondents evaluate training quality. Importantly, technology-related terms are not isolated; they co-occur with "đào tạo" and "chương trình," indicating that stakeholders conceptualize digital transformation as embedded

within curriculum design rather than as a peripheral add-on. The repeated pairing of "thực tiễn" and "thực hành" further suggests that respondents interpret digital readiness in applied rather than purely theoretical terms.

Substantively, the bigram structure deepens the quantitative findings. While Table 2 indicated moderately positive evaluations of curriculum (CT), the qualitative co-occurrence patterns reveal a persistent emphasis on practical applicability and technological updating. For instance, respondents frequently articulated that "chương trình cần cập nhật công nghệ mới" and "tăng cường thực hành gắn với doanh nghiệp." These expressions clarify that even when mean scores are relatively high, stakeholders still perceive a gap between theoretical content and market-oriented application. Thus, the bigram analysis supports and nuances the descriptive results: training is viewed positively overall, yet the dominant discourse centers on practical integration, digital tools, and alignment with labor market demand.

Figure 5. Thematic Distribution



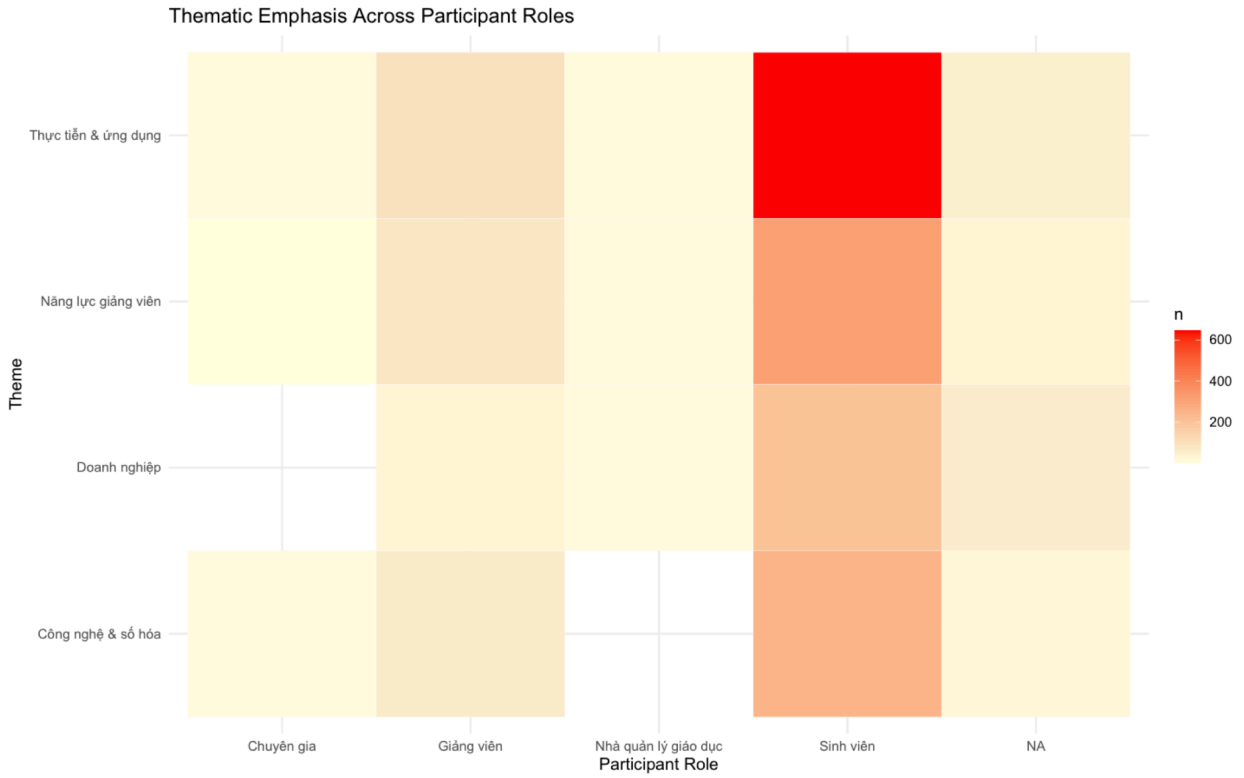
Source: Authors'compilation using R

**Notes: Figure aggregates coded responses into four thematic categories derived from axial coding: Thực tiễn & ứng dụng (Practical Application) Năng lực giảng viên (Lecturer Competence) Công nghệ & số hóa (Technology & Digitalization) Doanh nghiệp (Enterprise Linkage) Bars represent total coded frequency across all respondents. Themes reflect consolidated qualitative categories following saturation.*

Figure 5 aggregates the token-level data into four principal themes: (1) Thực tiễn & ứng dụng, (2) Năng lực giảng viên, (3) Công nghệ & số hóa, and (4) Doanh nghiệp. The distribution clearly indicates that "Thực tiễn & ứng dụng" is the most salient theme, followed by "Năng lực giảng viên," while "Công nghệ & số hóa" and "Doanh nghiệp" appear with slightly lower but still substantial frequencies. This hierarchy is analytically meaningful. It demonstrates that respondents do not treat digital transformation as an abstract technological agenda; rather, they frame it primarily through applicability and operationalization in real-world contexts.

These thematic results reinforce and extend the quantitative phase in several ways. First, the strong presence of the "Thực tiễn & ứng dụng" theme explains why curriculum-related items received moderately positive rather than extremely high evaluations in Table 2. Stakeholders appear to accept the overall structure but demand deeper integration of experiential learning components. Second, the salience of "Năng lực giảng viên" supports the relatively high mean scores for the lecturer construct (GV), while simultaneously suggesting expectations for continuous upgrading of digital pedagogical competence. Typical responses emphasized that "giảng viên cần nâng cao năng lực công nghệ số" and "đổi mới phương pháp giảng dạy theo hướng thực hành." Thus, the qualitative layer clarifies that strong evaluations of lecturers coexist with expectations of further digital capability development. The thematic hierarchy therefore validates and contextualizes the quantitative patterns rather than contradicting them.

Figure 6. Role-Specific Thematic Emphasis



Source: Authors'compilation using R

**Notes: Heatmap presents thematic intensity by stakeholder role (Students, Lecturers, Educational Managers, Experts). Color gradient represents frequency count of coded segments per theme within each group. Higher intensity indicates stronger thematic emphasis. Differences reflect role-based interpretive lenses rather than statistical testing.*

Figure 6 displays thematic intensity across stakeholder groups. Students demonstrate the highest concentration across all themes, particularly "Thực tiễn & ứng dụng" and "Năng lực giảng viên." Lecturers also emphasize these themes but show relatively stronger references to professional competence and institutional constraints. Educational managers display a more governance-oriented distribution, while experts' responses are comparatively strategic and macro-level in orientation.

This role-differentiated emphasis provides interpretive depth to the ANOVA results reported in Table 7. Quantitatively, significant differences emerged across roles for lecturer quality (GV), infrastructure (CS), and training management (QL), while

curriculum (CT) and strategic orientation (PH) showed no significant divergence. The qualitative heatmap clarifies why. Students' heavy emphasis on practical application and lecturer competence suggests an experience-based evaluation lens, whereas lecturers and managers frame issues in terms of implementation feasibility and structural constraints. For example, lecturers frequently noted "thiếu nguồn lực cập nhật công nghệ" and managers highlighted "rào cản cơ chế, tài chính." These narratives illuminate how institutional hierarchy and differential exposure to governance processes shape perception. The qualitative evidence therefore strengthens the interpretation that role-based differences are structural rather than attitudinal in nature.

4.5. Integration of Findings

The integration of quantitative and qualitative results reveals a coherent but multi-layered picture of training quality under digital transformation.

Quantitatively, stakeholders expressed generally positive evaluations across constructs, with particularly strong scores for lecturer competence (GV), development orientation (PH), and proposed solutions (GP). However, curriculum (CT) and infrastructure (CS) received comparatively lower means, and role-based differences were observed for governance-related dimensions.

Qualitatively, these patterns are meaningfully elaborated. The dominance of themes related to practical application and technological updating explains why curriculum evaluations, though positive, were not uniformly strong. Respondents consistently emphasized "gắn kết với doanh nghiệp," "tăng thực hành," and "cập nhật công nghệ," suggesting that structural modernization remains an ongoing process. Similarly, the high quantitative evaluation of lecturers is complemented by qualitative calls for enhanced digital pedagogy, indicating that competence is recognized but must evolve continuously.

Moreover, the absence of gender differences in the quantitative phase aligns with the qualitative data, where thematic emphasis did not vary meaningfully by gender but rather by institutional role. This convergence strengthens the interpretation that digital

readiness perceptions are shaped more by structural position than by demographic characteristics.

Taken together, the mixed-methods evidence indicates a higher education system characterized by strategic awareness, strong instructional commitment, and forward-looking orientation, yet facing operational challenges in curriculum modernization and infrastructural harmonization. The qualitative findings do not overturn the quantitative results; instead, they provide explanatory texture, clarify stakeholder expectations, and validate the structural interpretation of institutional readiness for the digital economy. This triangulated evidence enhances the robustness of the study and positions the findings within broader international discussions on digital transformation in higher education systems.